

Numerical investigation into strength and deformability of veined rock mass

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Abstract

Veined rock masses can be found in porphyry copper deposits like at the El Teniente mine in Chile. Rock masses with cemented joints or veins cannot be accurately represented using the methods commonly applied to predict the deformability and strength of jointed rock masses. Laboratory tests on veined rock andesite from the El Teniente mine have shown that the veins control the fracturing process and strength. In this work, the behavior of veined rock was studied using a numerical model, built in the finite element software ABAQUS. In this analysis, the rock mass was modeled as an assemblage of blocks bonded by layers, representing the intact rock and veins respectively. Uniaxial compression tests on veined core-size specimens were simulated and their results compared to experimental data. The results show good agreement with the experiments in terms of strength, stiffness and mode of failure. After this, a veined rock mass volume was modeled using the orientation of mapped and post-processed discontinuities as input. Simulations of uniaxial and triaxial tests were performed on the modeled veined rock mass. The observed failure mode confirms experimental results, showing that at low confinement the failure is concentrated in the veins and the rupture of intact rock becomes significant at higher confinement. The obtained strength envelope of the veined rock mass is non-linear and its shape is independent of the strength of the intact rock; however, the Hoek-Brown criterion did not fit the results. The residual strength of the veined rock approximated the residual strength of the intact rock, agreeing with previously reported observations in jointed rock.

Keywords Veined rock; Rock mass modeling; Numerical model; El Teniente mine; ABAQUS; Finite element method

1 Introduction

The characterization of the rock mass is a complex task due to its heterogeneous, discontinuous and anisotropic character. The mechanical properties of the rock mass are often governed by the presence of discontinuities. Field testing can provide more accurate information, but for several reasons it is often difficult to perform. Because of this, the behavior of the rock mass is often assessed through experience, classifications and characterization systems (e.g. RMR, RMi, Q, GSI) and empirical approaches.¹⁻⁴

The behavior of the rock mass can also be estimated using numerical methods. Given the discontinuous nature of the rock mass, methods capable of explicitly including the rock fractures offer the advantage of studying the complex interactions between intact rock blocks and joints during loading. Numerical methods allow modeling large rock volumes. They also offer the capability to model rock masses with cemented joints or veins, which cannot be accurately represented in classifications and characterization systems commonly used to predict the deformability and strength of jointed rock masses.

Veined rock masses can be found in porphyry copper deposits like at the El Teniente mine in Chile. The El Teniente mine is an underground copper mine located in the central Andes of Chile and

the largest known copper-molybdenum deposit in the world.⁵ The rock masses at the El Teniente mine can be described as very competent and massive, exhibiting brittle behavior.^{6,7} The main discontinuity types observed within the primary copper ore at El Teniente are widely spaced faults and a stockwork composed by a dense network of thin veins. The veins are healed or cemented, principally with quartz, sulfides and anhydrite, and are characterized by intermediate to high strength.^{7,8} It has been observed that the rock mass detachment and fragmentation during caving propagation occurs mainly through the veins.⁸

The high tensile and shear strength of mineral veins compared to joints can significantly affect the rock mass behavior. Bend tests on shale containing calcite-filled veins have shown that the geometric properties and thickness of the veins have a strong influence on fracture propagation.⁹ Results of triaxial compression tests on veined andesite from the El Teniente mine have demonstrated that the fracture of the rock is controlled by the veins and that their strength determines the overall strength behavior.^{10,11,12} The rupture of veined rock tends to be controlled by critically oriented veins.¹⁰

Experimental studies on strength of veined rock are scarce and only a few can be found in the literature. It is difficult to isolate and understand the effect of the different vein parameters on the behavior of rock because of the difficulty in reproducing the tests.¹⁴ Given this limitation, a good alternative is to study the mechanical behavior of veined rock using numerical methods.

Numerical modeling using the discrete element method (DEM) has been applied to study the fracture behavior of veined rocks. Investigations using the particle flow code (PFC) and the synthetic rock mass methodology (SRM) have simulated laboratory tests on veined rock specimens.¹³ These studies show the strong effect of vein strength on the overall mechanical strength.^{10,12-16} Fractures are localized within weak veins or they initiate along the interface between rock and strong veins and propagate depending on vein orientation.¹⁶ These kind of models can fairly reproduce the behavior of veined core-size specimens in uniaxial and triaxial compression.^{10,12,17} However, although this method has a high potential, it requires a laborious calibration process and large computation time among other limitations.

In the present work, the finite element program ABAQUS¹⁸ was used to study the behavior of veined rock mass. The experimental work reported by Suzuki,¹⁷ Vallejos et al.¹² and Jacobsson et al.¹⁹ was used as a basis for the validation of the numerical model.

The rock mass was modeled as an assemblage of intact rock blocks, divided and bonded by thin layers representing the veins. The advantage of this method is that the input parameters for the intact rock and discontinuities can be determined directly from laboratory tests. First, the applicability of the so-called cohesive elements for the representation of the discontinuity planes in the veined rock was examined. After this, the results of simulations of uniaxial tests performed on veined core-size specimens were compared with experimental results. Finally, the behavior of a larger rock mass block was studied by simulating uniaxial and triaxial tests.

2 Experimental data

The behavior of the modeled veined rock mass was verified by comparing the results of numerical simulations with experimental results. The available experimental work corresponds to direct shear and uniaxial tests, performed on veins and veined core-size specimens of the El Teniente Mafic Complex (CMET) lithology, reported by Suzuki,¹⁷ Vallejos et al.¹² and Jacobsson et al.¹⁹

The cylindrical specimens tested in uniaxial compression had an aspect ratio (H/D) of 2:1, with a diameter of 50 mm. The specimens were tested under load control and the deformation was measured using strain gauges. Pictures of four of the ten specimens tested are presented in Fig. 1.

The veins in the specimens were mapped and examined before and after testing. The vein mapping considered veins with thickness greater than 0.3 mm. The composition of the veins infill was

found to be a combination of two or more minerals such as anhydrite, quartz, chalcopyrite, bornite, molybdenite, biotite, chlorite or pyrite, being the first three the most predominant.¹⁷ The infill of each vein was reported according to its most abundant mineral. In the uniaxial tests, the specimens failed mostly through quartz and anhydrite veins.



Figure 1. Examples of veined specimens after uniaxial test.^{12,17}

The mechanical parameters of the veins were estimated from triaxial tests on veined specimens and direct shear, direct tension and indirect tension tests performed on rock veins.^{17,19} The mechanical parameters of the intact rock of the CMET were obtained from different series of uniaxial compression tests on samples that failed through the intact rock.^{12,17}

Ranges for the values of elastic and strength parameters of the veins were estimated depending on the most predominant mineral present in the vein. A detailed analysis of the test results is provided by Suzuki.¹⁷ The normal stiffness of the veins was found to be lower than the shear stiffness, which was attributed to problems with the testing equipment.¹² A summary of the properties of intact rock and veins obtained from the experiments and considered in this study is presented in Tables 1 and 2.

Table 1. Mechanical parameters of intact rock of the CMET lithology.^{17,20}

Material property	Value
Intact rock Young's modulus, E_i (GPa)	55
Intact rock Poisson's ratio, ν	0.25
Intact rock unconfined compressive strength σ_c (MPa)	120
Intact rock Hoek-Brown parameter m_i	12

Table 2. Mechanical parameters of veins of the CMET lithology.^{12,17}

Material property	Vein mineral predominant composition		
	Quartz (qz)	Chalcopyrite (cpy)	Anhydrite (anh)
Joint shear stiffness, k_s (GPa/m)	17.0 - 116.2	47.1 - 52.6	25.6 - 129.1
Joint cohesion, c_{jp} (MPa)	42	11 - 19	26 - 34
Joint angle of friction, ϕ_{jp} ($^{\circ}$)	34	38 - 43	31 - 39
Joint tensile strength, σ_{jt} (MPa)	4.3	1.3	1.6

3 Description of the numerical model

3.1 Geometry

The model geometry was generated using a code implemented in MATLAB²¹. Within the code, the modeling volume is defined as a parallelepiped intersected by planes. The dimensions of the volume, the number of planes contained in the block, their orientation and their thickness can be specified. The planes can be generated individually or as sets with specified spacing and orientation, including statistical variation. The geometry created in MATLAB can be exported in the form of a script written in Python programming language, which is used by ABAQUS CAE to generate the geometry of the model. The resulting ABAQUS model consists of blocks representing the intact rock and thin layers representing discontinuities. The thickness of the layers can be specified.

The modeled volume can be reshaped in ABAQUS CAE by intersecting it with another volume of a different shape. This is done, for example, to obtain cylindrical specimens. An example of a model generated with the MATLAB code and the resulting geometry in ABAQUS is shown in Fig. 2. After the generation of the geometry in ABAQUS is completed, the model is meshed and the element types are assigned.

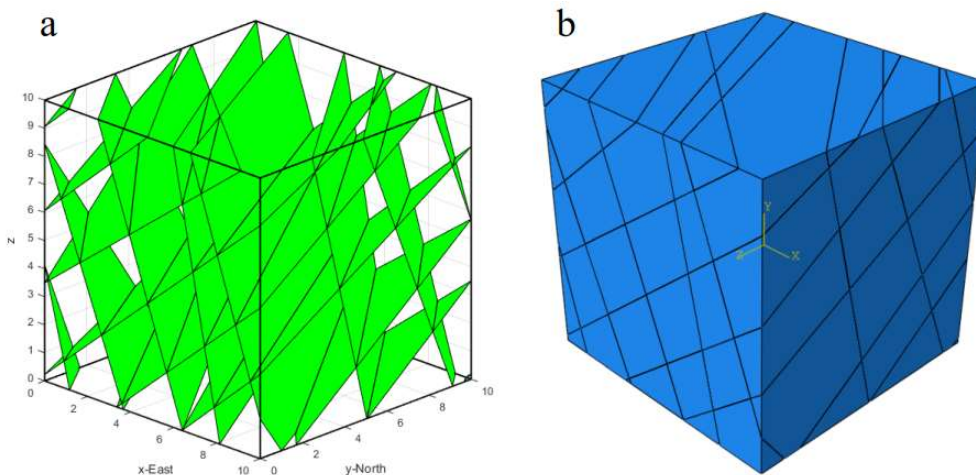


Figure 3. a) Example of geometry with two discontinuity systems generated using MATLAB script. b) Model geometry in ABAQUS.

3.2 Solid and contact elements

In the modeled veined rock mass, consisting of intact rock and discontinuities, the intact rock was represented as solid elements. Their mechanical behavior was defined as elasto-plastic and it was controlled through a user defined subroutine (VUMAT).

The discontinuities (veins) were modeled using cohesive elements. The cohesive elements, included in the ABAQUS library, are conceived for the modeling of adhesives, gaskets or in situations where the integrity of the interface is of importance.

The cohesive elements were included as a layer in the interface between two solid rock blocks. The contacts bonding the cohesive element layers to the solid elements were defined as a *tie constraint*. The mechanical behavior of the cohesive elements was controlled with a user defined subroutine (VUMAT). In this subroutine, the contact elements were used in *traction-separation*, assuming uncoupled behavior between normal and tangential stresses as shown in Eq. 1.

$$t = \begin{Bmatrix} t_n \\ t_s \\ t_t \end{Bmatrix} = \begin{bmatrix} E_{nn} & 0 & 0 \\ 0 & E_{ss} & 0 \\ 0 & 0 & E_{tt} \end{bmatrix} \begin{Bmatrix} \varepsilon_n \\ \varepsilon_s \\ \varepsilon_t \end{Bmatrix} = E \varepsilon \quad (1)$$

Here, t corresponds to the stress, E to the stiffness and ε to the strain. The indices n , s and t represent the normal and two shearing directions respectively.

The mechanical behavior of the contact elements, in each of the three orthogonal directions, was assumed as linear elastic with a strength limit. For the simulations performed within this work, the normal behavior was assumed linear elastic with tension cut-off. The shear behavior was assumed as elasto-plastic with Mohr-Coulomb failure criterion (Eq. 2), in which the shear strength τ is a function of the normal stress σ , the cohesion c and friction angle φ .

$$\tau = c + \sigma \tan \varphi \quad (2)$$

4 Verification of the discontinuity model

The performance of the cohesive elements and the constitutive model to simulate the vein behavior was verified. The results of numerical simulations were compared with the results of direct shear and direct tensile tests performed on rock veins from the El Teniente mine.¹⁹

The model for the simulation of the direct shear test consisted of three components: a lower block, an upper block and a thin layer between the blocks to simulate the discontinuity. Each block had dimensions of 30 x 30cm with 7.5 cm height and the cohesive element layer had a thickness of 1 mm. The blocks were defined as elastic material. The layer between the blocks was modeled using 8-nodes cohesive elements.

An example of a direct shear test under normal stresses of $\sigma_n = 5$ and 10 MPa is shown in Fig. 3. The values of the parameters used in this simulation are listed in Table 3. To reproduce the strength observed in the shear tests, a softening process was implemented by linearly reducing the parameters of the peak shear strength until the residual strength was reached. An approximation to the strength and strain of the vein was achieved by using twice the failure shear strain to reach the residual strength (Fig. 3a).

The direct tension test was modeled using the same geometry and by pulling the upper block while holding the lower block. The results of the test and the experimental result are shown in Fig. 3b.

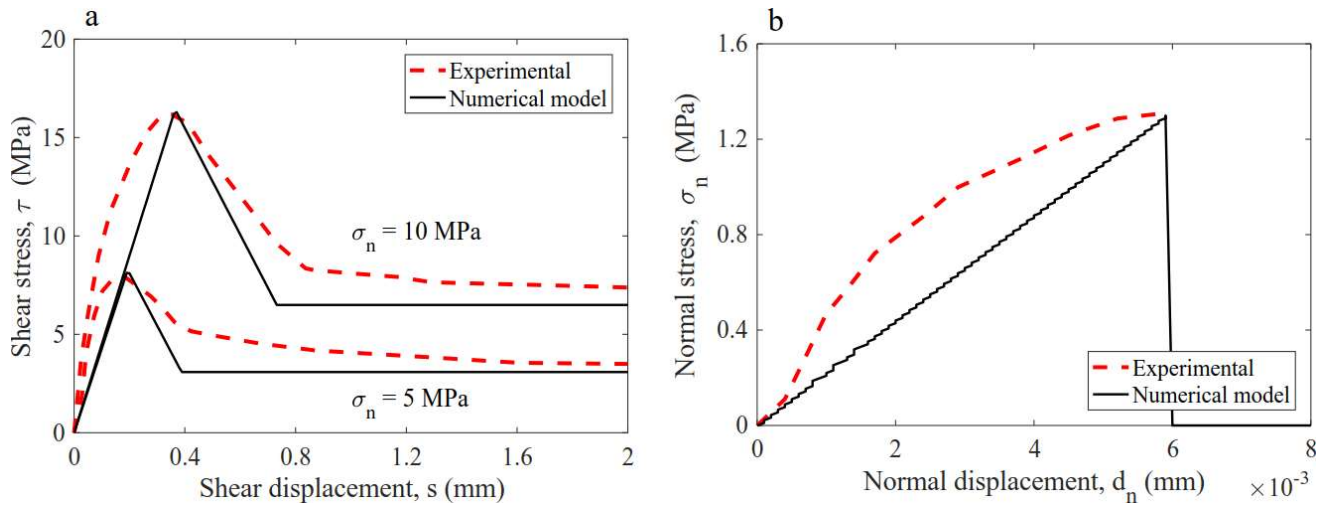


Figure 3. Comparison between simulation and laboratory test.¹⁹ a) Example of a direct shear tests under normal stress of $\sigma_n = 5$ and 10 MPa. b) Example of a direct tension test.

As observed in Fig. 3a, the results of the shear test show an initial higher stiffness and lower final stiffness when compared to the numerical model. The behavior of the contacts has been assumed as linear elastic with a strength limit. This simplification was necessary given the description of the specimens and the available experimental results. Additional experimental data would be necessary to obtain more parameters that allow improving the performance of the numerical model. The limitations of the chosen contact model make difficult to reproduce exactly the behavior observed in the experiments. Nevertheless, the numerical model can reproduce the strength and deformation of rock veins in a satisfactory manner.

Table 3. Mechanical parameters used for the simulation of the direct shear and direct tension tests (Fig. 3).¹⁹

Material property	Value
Intact rock Young's modulus, E_i (GPa)	55
Intact rock Poisson's ratio, ν	0.25
Joint normal stiffness (secant), k_n (GPa/m)	220
Joint shear stiffness (secant), k_s (GPa/m)	45
Joint peak cohesion, c_{jp} (MPa)	1
Joint angle of friction, ϕ_{jp} (°)	57
Joint residual angle of friction, ϕ_{jr} (°)	33
Joint tensile strength, σ_{jt} (MPa)	1.3

5 Uniaxial compression tests on veined core-size specimens - Comparison of numerical simulations with experimental data

A total of four uniaxial tests were simulated. Each numerical specimen with its vein network was created using the methodology described in Section 3.1. The veins were modeled using cohesive elements as described in Section 4. The intact rock was modeled using second-order tetrahedral elements.

Numerical models of four veined specimens were created: NNM16-1, NNM7-4 NNM21-1 and NNM22-1.^{12,17} The first three specimens were selected because they are well described in the

experimental work¹⁷ and their test results were fairly good reproduced in the numerical modeling using PFC.¹² Thus, it was assumed that the description of the vein network and vein minerals is probably more accurate than in other cases. Specimen NNM22-1 was included for no particular reason, other than to have an additional test for comparison.

5.1 Intact rock properties

According to the experimental results, the failure of the veined specimens in uniaxial compression occurred along their discontinuities. This can be expected given the considerable number of discontinuities and the high strength of the intact rock. However, in some cases, partial failure through the intact rock was observed.

The intact rock was modeled as elasto-plastic material with the Hoek-Brown failure criterion for rock masses,²² expressed as:

$$\begin{aligned}\sigma_1 &= \sigma_3 + \sigma_c \left(\frac{m_b \sigma_3}{\sigma_c} + s \right)^a \\ m_b &= m_i \exp \left(\frac{GSI-1}{28-14D} \right) \\ s &= \exp \left(\frac{GSI-100}{9-3D} \right) \\ a &= \frac{1}{2} + \frac{1}{6} \left(\exp \left(\frac{-GSI}{15} \right) - \exp \left(\frac{-20}{3} \right) \right)\end{aligned}\tag{3}$$

where σ_c is the uniaxial compressive strength (UCS) and m_i is a dimensionless empirical parameter that depends on the intact rock properties. The parameter D accounts for blast damage and relaxation. With these equations, the strength of the rock mass is calculated depending on the rock characterization given by the Geological Strength Index (GSI).

The properties of the intact rock used in the simulation are those presented in Table 1. The intact rock behavior was assumed as brittle. For the peak strength of the intact rock (GSI=100), the parameters in Eq. 3 result: $m_b=m_i$, $s=1$ and $a=0.5$. The residual parameters m_{br} , s_r and a_r were calculated assuming GSI=40, which was found to characterize the residual strength of intact granite specimens tested in triaxial compression.²³ The parameter D was always assumed as zero.

The constitutive model for the intact rock was implemented into an ABAQUS user subroutine (VUMAT).

5.2 Vein properties

The behavior of the veins was assumed as elasto-plastic, with Mohr-Coulomb failure criterion and tension cut-off as described in Section 3.2.

As explained in Section 2, ranges for the values of elastic and strength parameters of the veins were estimated depending on their mineralogy (Table 2). Values within these ranges were tried, in order to find those that reproduce a behavior similar as that observed in the uniaxial tests. For a given vein mineral, the same parameter values were assumed in all four simulations of uniaxial tests. The adopted properties are presented in Table 4.

The normal stiffness was estimated from the obtained shear stiffness k_s as $k_n = 3k_s$. This ratio agrees with values used to model glued fractures²⁴ and is compatible with the elasticity theory, since the discontinuities are filled and cemented. The dilation angle in the contacts was set to 0°, assuming that most of the dilation is produced by the rotation of blocks during loading and failure.

Table 4. Mechanical parameters of discontinuities used in the modeled core-size specimens

Property	Vein mineral predominant composition		
	Quartz (qz)	Chalcopyrite (cpy)	Anhydrite (anh)
Joint normal stiffness (estimated), k_n (GPa/m)	100	132	150
Joint shear stiffness, k_s (GPa/m)	33	44	50
Joint peak cohesion, c_{jp} (MPa)	42	19	32
Joint residual cohesion, c_{jr} (MPa)	0	0	0
Joint peak angle of friction, ϕ_{jp} (°)	39	40	41
Joint residual angle of friction, ϕ_{jr} (°)	39	40	41
Joint tensile strength, σ_{jt} (MPa)	4.3	1.3	1.6

5.3 Vein network

The orientations of the veins in the specimens were obtained from maps of the specimens' surface.¹⁷ The maps of their vein network, and included in the model, are shown in Fig. 4. The orientation and the main mineral present in the vein are indicated. The veins of the tested specimens had thicknesses ranging between 0.3 and 2 mm. In the model, a thickness of 1 mm was assumed for all veins. The discontinuity intensity in the modeled specimens, expressed as the ratio between the total discontinuity area and the specimen's volume, is given in Table 5. The modeled specimens are shown in Fig. 5.

Table 5. Discontinuity intensity in modeled core-size specimens.

Model	Volume (cm ³)	Vein area (cm ²)	Vein intensity (cm ² /cm ³)
NNM7-4	196	253	1.29
NNM16-1	196	226	1.15
NNM21-1	196	327	1.67
NNM22-1	196	282	1.43

5.4 Results

The uniaxial tests were simulated by applying a constant vertical deformation at the top, while preventing the vertical displacement at the base. The axial strain was calculated as the ratio between the displacement at the top and the initial height of the specimen. The lateral (radial) strain was calculated from the displacements in the middle portion of the specimen.

5.4.1 Stress-strain relation

Fig. 6 shows the stress-strain curves obtained from the three modeled specimens and from the experimental results. Since the laboratory tests were performed using load control, their post peak behavior could not be recorded.

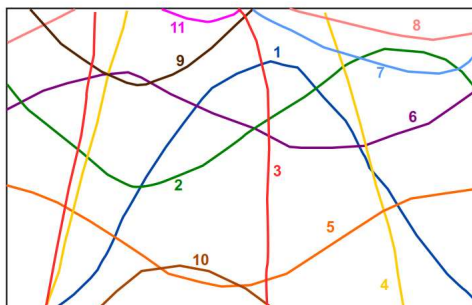
It is observed that there is somehow good agreement between experimental and simulated stress-strain curves. The numerical model reproduces the peak strength and strain obtained in the experiments. The failure is brittle, and the stress drops to zero after reaching the peak strength. However, the experimental curves show the typical characteristics of compression tests, such as initial hardening due to crack closure and softening due to crack initiation. These aspects are not reproduced

by the numerical model, probably because of the simplicity of the constitutive models used to represent the intact rock and veins.

The values of the Young's modulus, Poisson's ratio, uniaxial compressive strength and failure strain obtained from the numerical model are given in Table 6. These values agree well with the experimental results. Differences between experiments and numerical simulation could also be caused by the measurements of the deformation: the axial deformation in the experiments was measured locally, whereas in the model this was calculated along the entire length of the specimen. It must be pointed out that local measurement (strain gauge) of the deformation in a jointed specimen can lead to error in the estimation of the overall behavior.

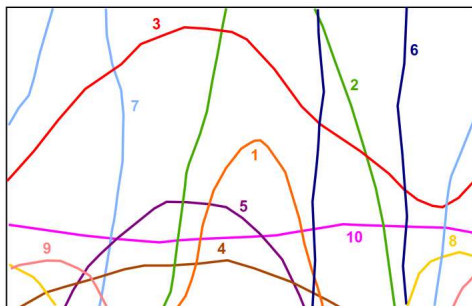
NNM7-4

Vein	Dip (°)	Dip direction (°)	Main mineral
1	60	270	qz
2	44	180	anh
3	83	348	qz
4	78	304	qz
5	35	124	qz
6	28	41	cpy
7	30	338	qz
8	20	332	qz
9	52	188	anh
10	46	335	qz
11	45	165	qz



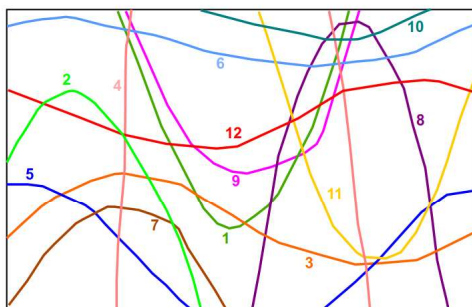
NNM16-1

Vein	Dip (°)	Dip direction (°)	Main mineral
1	78	60	anh
2	82	40	cpy
3	54	90	cpy
4	25	91	qz
5	55	90	qz
6	89	160	qz
7	89	200	anh
8	57	248	qz
9	55	200	qz
10	5	250	qz



NNM21-1

Vein	Dip (°)	Dip direction (°)	Main mineral
1	72	360	qz
2	66	300	anh
3	31	253	qz
4	82	179	qz
5	86	213	qz
6	45	332	qz
7	19	289	qz
8	60	256	qz
9	78	90	cpy
10	67	352	qz
11	19	280	anh
12	72	240	qz



NNM22-1

Vein	Dip (°)	Dip direction (°)	Main mineral
1	50	21	qz
2	50	250	cpy
3	75	58	qz
4	78	107	qz
5	10	311	qz
6	62	245	anh
7	50	175	anh
8	40	243	anh
9	15	225	qz

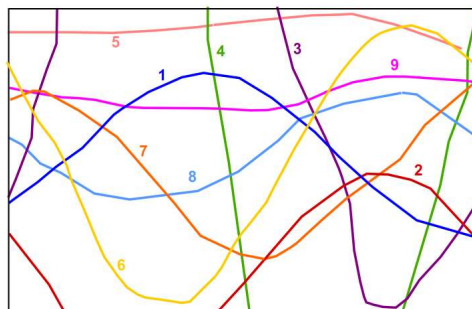


Figure 4. Vein network in specimens NNM7-4, NNM16-1, NNM21-1 and NNM22-1. anh: Anhydrite. cpy: Chalcopyrite. qz: Quartz.^{12,17}

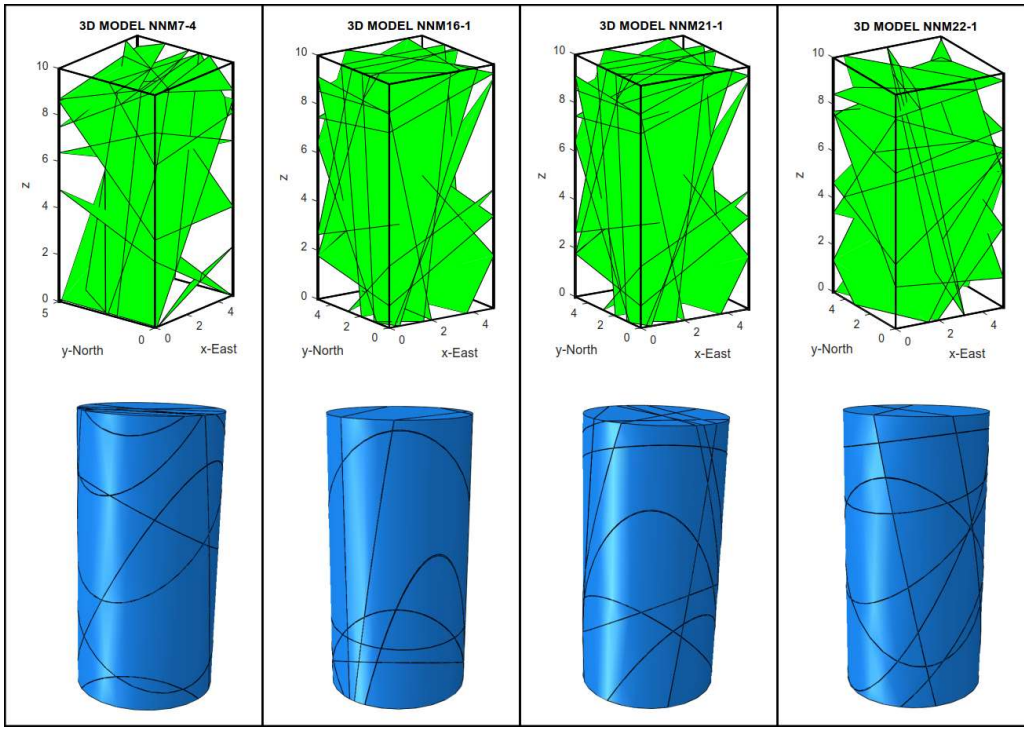


Figure 5. Numerical models generated using MATLAB script (top) and models in ABAQUS (bottom): NNM7-4, NNM16-1, NNM21-1 and NNM22-1.

Table 6. Comparison of strength, elastic parameters and failed veins in numerical models of core-size specimens and laboratory tests.

Specimen		σ_{cm} (MPa)	E_m (GPa)	ν	Vein in failure
NNM7-4	Experimental	117	29	0.27	2 (anh), 3 (qz) and 4 (qz)
	Numerical model	121	34	0.29	1 (qz), 3 (qz) and 4 (qz)
NNM16-1	Experimental	71	44	0.16	2 (cpy), 3 (cpy) and 6 (qz)
	Numerical model	72	47	0.24	2 (cpy), 3 (cpy) and 7 (anh)
NNM21-1	Experimental	100	37	0.26	1 (qz), 4 (anh) and 8 (qz)
	Numerical model	102	40	0.23	1 (qz) and 8 (qz)
NNM22-1	Experimental	99	37	0.25	3 (qz), 4 (qz) and 6 (anh)
	Numerical model	99	40	0.27	3 (qz), 6 (anh) and 7 (anh)

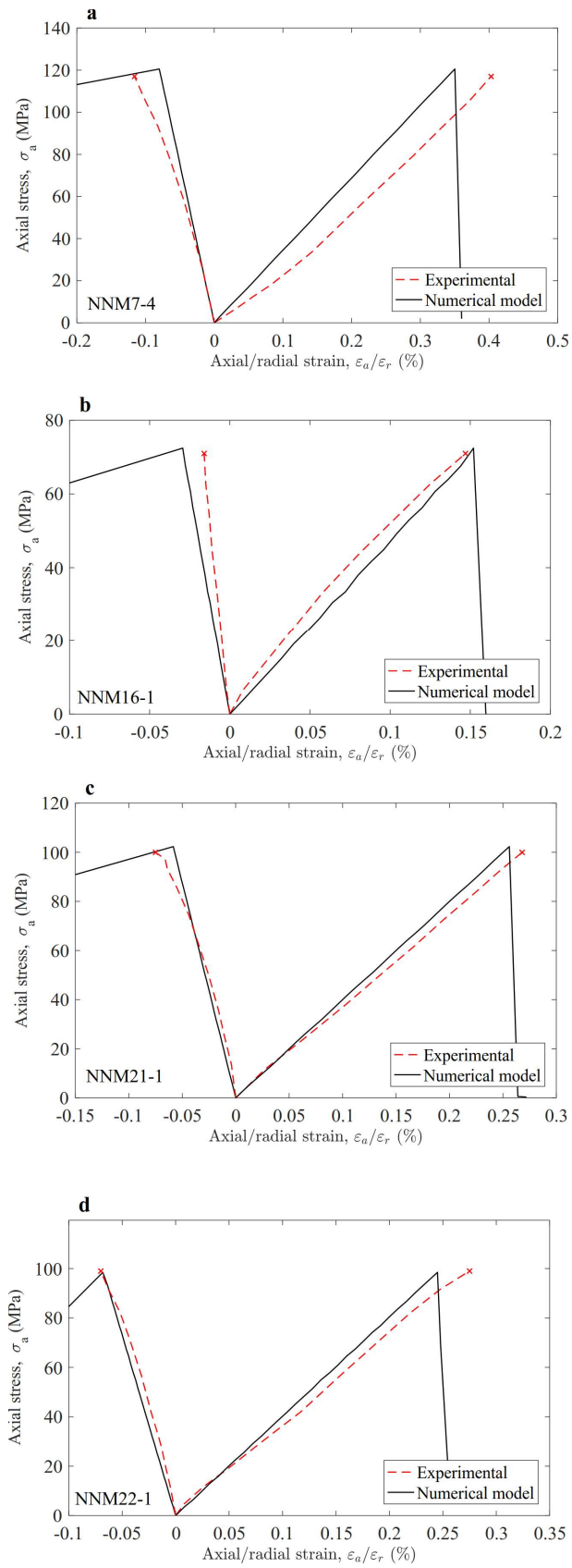


Figure 6. Comparison between stress-strain curves from laboratory tests and numerical models.

5.4.2 Failure mode

In the numerical model, the mode of failure was determined by analyzing the failure of the veins at peak strength. The type of failure, tensile or shear, was saved in the model using a state variable. In addition, the contours of deformation indicated the veins involved in the failure. Examples of the deformed specimens after reaching the peak strength are shown in Fig. 7.

The specimens had mostly persistent veins and the failure occurred mainly through them. Even if non-persistent veins could be included, they cannot be automatically inserted into the geometry generated by the MATLAB code. For this reason, these were not included into the models. This assumption can affect the results and the observed failure mode.

The results of the simulations show certain agreement with the experimental observations. In the laboratory test, the specimen NNM7-4 failed mainly through the vein 2 (anhydrite), 3 and 4 (quartz). In the numerical model, the failure occurred principally through veins 1, 3 and 4 (quartz). The vein 1 (quartz) failed mainly through shear, whereas veins 3 and 4 presented partial tensile failure.

According to the experimental work, the specimen NNM16-1 failed mainly through veins 2 and 3 (chalcopyrite) and through the vein 6 (quartz). In the numerical model, the failure occurred mainly through the veins 2, 3 and partially also 7 (anhydrite), which has similar orientation to vein 6.

Specimen NNM21-1 failed through veins 1 (quartz), 4 (anhydrite) and 8 (quartz) in the laboratory. In the numerical model, the failure occurred mainly through veins 1 and 8.

Specimen NNM22-1 failed mainly through the intact rock and partially through the veins 3 (quartz), 4 (quartz) and 6 (anhydrite). In the numerical model, the principal failure of the specimen was through the same veins 3, 6 and 7, which failed mainly in tension.

In general, giving the simplicity of the material behavior assumed for intact rock and veins, there is a good agreement between the experimental results and the simulations in terms of strength, elastic properties and failure mode.

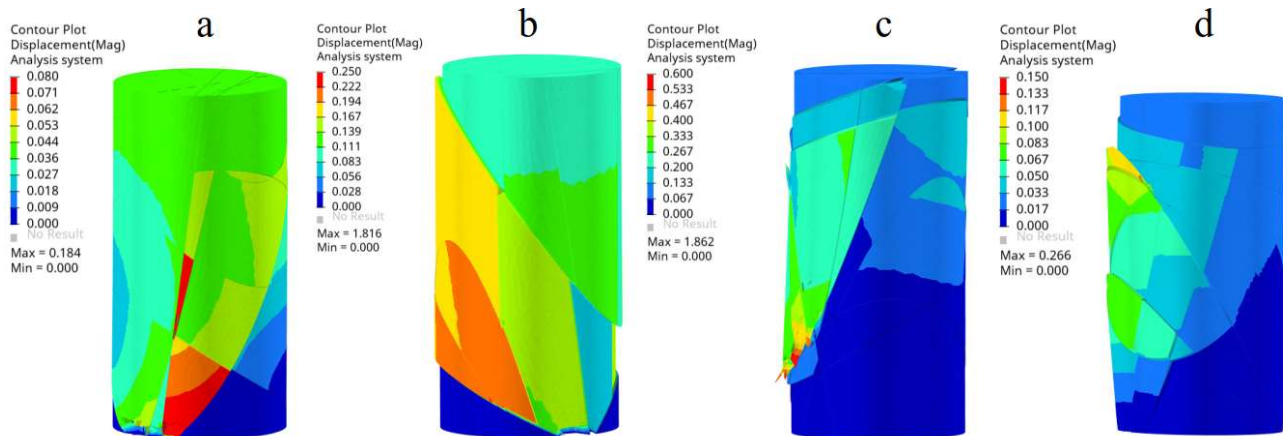


Figure 7. Total displacement in modeled specimens NNM7-4 (a), NNM16-1 (b), NNM21-1 (c) and NNM22-1 (d).

6 Rock mass modeling

In the previous section, it has been shown that the numerical model reproduces satisfactorily the strength and stiffness of the core-size veined specimens. Using the same methodology, a numerical model was created to study the rock mass behavior at El Teniente mine.

A rock mass model of the CMET lithology was built. The modeled rock mass volume corresponds to an 8 m³ veined rock block, a size that would be adequate for the analysis of the rock mass behavior at the tunnel scale. The behavior of the veined rock mass was studied through simulations of uniaxial and triaxial tests.

6.1 Discontinuity network

Information about orientation, position and the equivalent diameter of the discontinuity sets was obtained from surface trace mappings and post-processing with the discrete fracture network (DFN) software FracMan.^{7,25} The orientations of the considered veins are presented in Table 7 and Fig. 8a.

The volumetric discontinuity density (intensity), calculated as the ratio between the total vein area and the rock mass volume, results 8.2 m²/m³ approximately. From the total number of discontinuities, about 20 % of them correspond to the sub-horizontal system 1. The systems 2 (66/262) and 5 (69/076) are also relevant, with seven and six veins and an intensity of about 1.8 m²/m³ and 1.4 m²/m³ respectively. The intensity of the remaining systems varies between 0.05 m²/m³ and 0.6 m²/m³.

Systems with intensities smaller than 0.1 m²/m³ were not included in the numerical model. The geometry was generated using the vein orientation as input for the MATLAB code mentioned in Section 3. The vein systems included in the model are given in Table 8 and shown in Fig. 8b. The discontinuity intensities of the input and processed vein systems are compared in Table 9. The resulting geometry of the modeled rock mass volume is shown in Fig. 9.

Table 7 Orientation and properties of the vein sets obtained from mapping and post-processing (Fig. 8).

System	Number of veins	Percentage of poles (%)	Dip (°)	Dip direction (°)
1	11	21.2	5±4	180 ±96
2	7	13.5	66 ±4	262 ±7
3	6	11.5	35 ±5	250 ±12
4	3	5.8	15 ±1	268 ±4
5	6	11.5	69 ±8	76 ±7
6	4	7.7	40 ±8	82 ±6
7	3	5.8	43 ±6	201 ±8
8	3	5.8	24 ±2	12 ±7
9	4	7.7	57 ±4	39 ±9
10	2	3.8	85 ±2	57
11	1	1.9	54	179
12	1	1.9	61	106
13	1	1.9	85	40

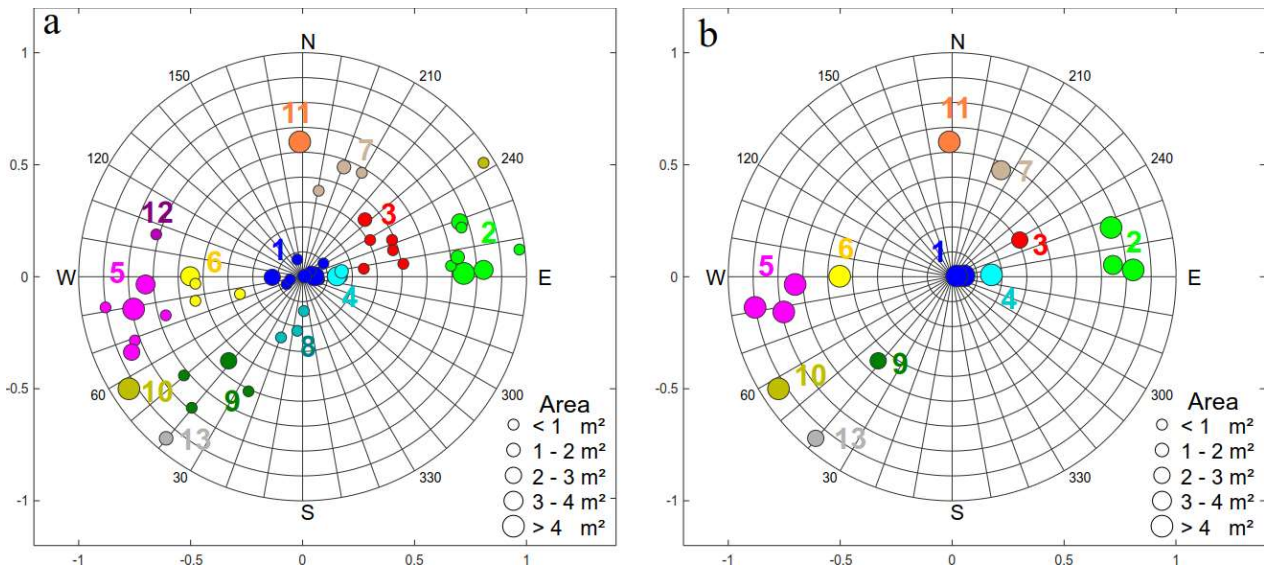


Figure 8. a) Stereographic projection of the vein sets obtained from surface mappings and post-processing with the DFN software FracMan. b) Vein sets included in the numerical model

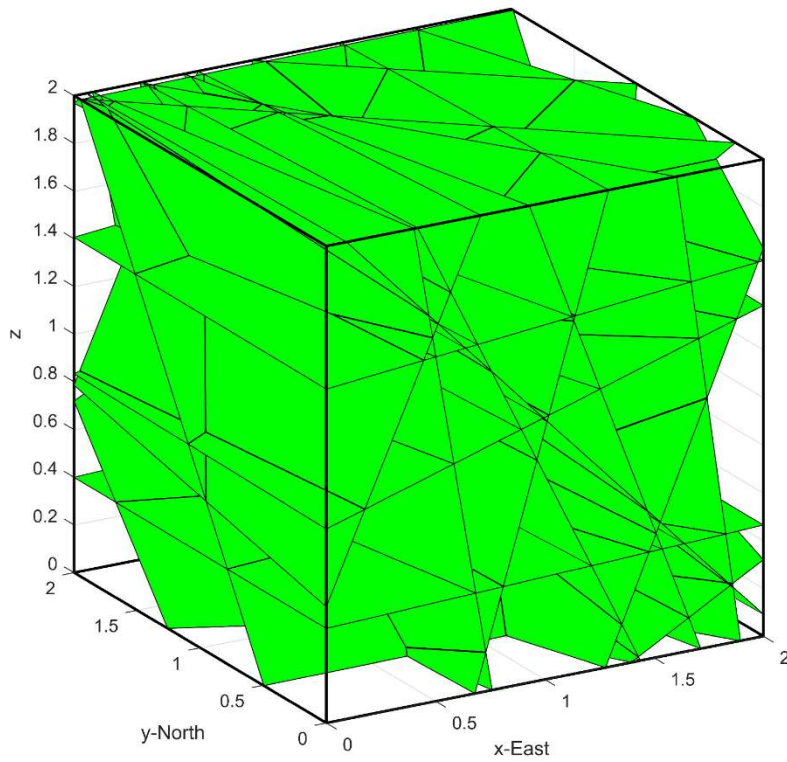


Figure 9. Numerical model geometry generated using MATLAB script

Table 8. Orientation of the vein sets included in the numerical model

System	Dip (°)	Dip direction (°)
1	5	268
	2	267
2	65	265
	73	268
	67	253
3	31	242
4	16	268
5	80	81
	69	78
	63	87
6	45	90
7	47	205
9	45	41
10	83	57
11	54	179
13	85	40

Table 9. Discontinuity intensity of input data and vein sets included in the rock mass numerical model.

System	Input data		Model	
	Area (m ²)	Intensity (m ² /m ³)	Area (m ²)	Intensity (m ² /m ³)
1	10.2	1.3	8	1
2	14.7	1.8	11.3	1.5
3	2.7	0.3	2.7	0.3
4	4.7	0.6	4.2	0.5
5	11.2	1.4	13	1.6
6	4.3	0.5	5.4	0.7
7	2.7	0.3	3.5	0.4
8	0.9	0.1	-	-
9	3	0.4	2.6	0.3
10	4.8	0.6	4.8	0.6
11	4.5	0.6	4.9	0.6
12	0.4	0.1	-	-
13	1.4	0.2	2.8	0.3
Total	65.5	8.2	63.2	7.9

6.2 Material properties

The intact rock was modeled in the same manner as explained in Section 4. The same mechanical parameters were used.

As explained in Section 2, the veins in the specimens of the CMET lithology were composed of different minerals mixed in different proportions.^{12,17} Quartz was the main mineral present in the veins with a proportion of 69%. The proportion of anhydrite and chalcopryrite was approximately 25% and 6% respectively.

In the numerical model, a unique type of vein was included. Its strength parameters and stiffness were calculated as the weighted average of the vein properties presented in Table 4, considering the mentioned proportions. The mechanical parameters of the intact rock and veins are given in Table 10.

Table 10. Mechanical parameters of intact rock and veins in the rock mass numerical model

Material property	Value
Intact rock Young's modulus, E (GPa)	55
Intact rock Poisson's ratio, ν	0.25
Intact rock density, γ (kN/m ³)	28
Intact rock uniaxial strength, σ_c (MPa)	120
Intact rock parameter m_i	12
Residual GSI of intact rock	40
Joint normal stiffness, k_n (GPa/m)	114
Joint shear stiffness, k_s (GPa/m)	38
Joint peak angle of friction, ϕ_{jp} (°)	40
Joint residual angle of friction, ϕ_{jr} (°)	40
Joint peak cohesion, c_{jp} (MPa)	38
Joint residual cohesion, c_{jr} (MPa)	0
Joint tensile strength, σ_{jt} (MPa)	3.4

6.3 Results of tests simulations

Triaxial tests were performed by first, applying an initial stress, equal to the confining stress, on all elements. The same stress was applied on the boundaries of the block, loading the specimen uniformly. The confining stress remained constant during the simulation. After this, the top face of the block was moved downwards with constant strain rate, while the displacement of the bottom face was restricted in the vertical direction. Uniaxial and triaxial tests with confining stress between 1 and 60 MPa were simulated.

6.3.1 Stress-strain relation

The stress-strain curves obtained from the simulations are shown in Fig. 10a. The curves show peak and residual strength. In uniaxial compression, the post-peak behavior shows softening and residual strength, compared to the brittle failure observed in the simulations of the specimens presented in Section 4. This can be caused by a relative larger number of fractures in the rock mass model. At low confining stress, the behavior is brittle, becoming more ductile with increasing confinement. This behavior is influenced by the failure mode.

The lateral strain is shown in Fig. 10b. The lateral strain was calculated from the deformation of the four lateral faces. At low confining stress, the lateral strain is lower in North-South than in the East-West direction (Fig. 10c). The deformation in both horizontal axes becomes similar with increasing confining stress.

The volumetric strain was calculated as the sum of the axial and lateral strains. The curves of volumetric vs. axial strain are shown in Fig. 11, indicating the axial strain at peak strength. The plots show compaction of the specimen during loading, and after failure, the specimens underwent dilatant behavior.

In the uniaxial test and at low confining stress, the specimen reached the peak strength after the onset of dilatancy. With increasing confinement, both, the onset of dilatancy and the peak strength occur at almost the same axial strain. As the behavior becomes more ductile ($\sigma_3 \geq 40$ MPa), the dilation

begins after the stress drops to the residual strength. Reported results of triaxial tests on veined andesite specimens show a similar behavior, in which the onset of dilatancy occurs short before reaching the peak strength.⁹ This means a short stage of unstable fracturing before failure.

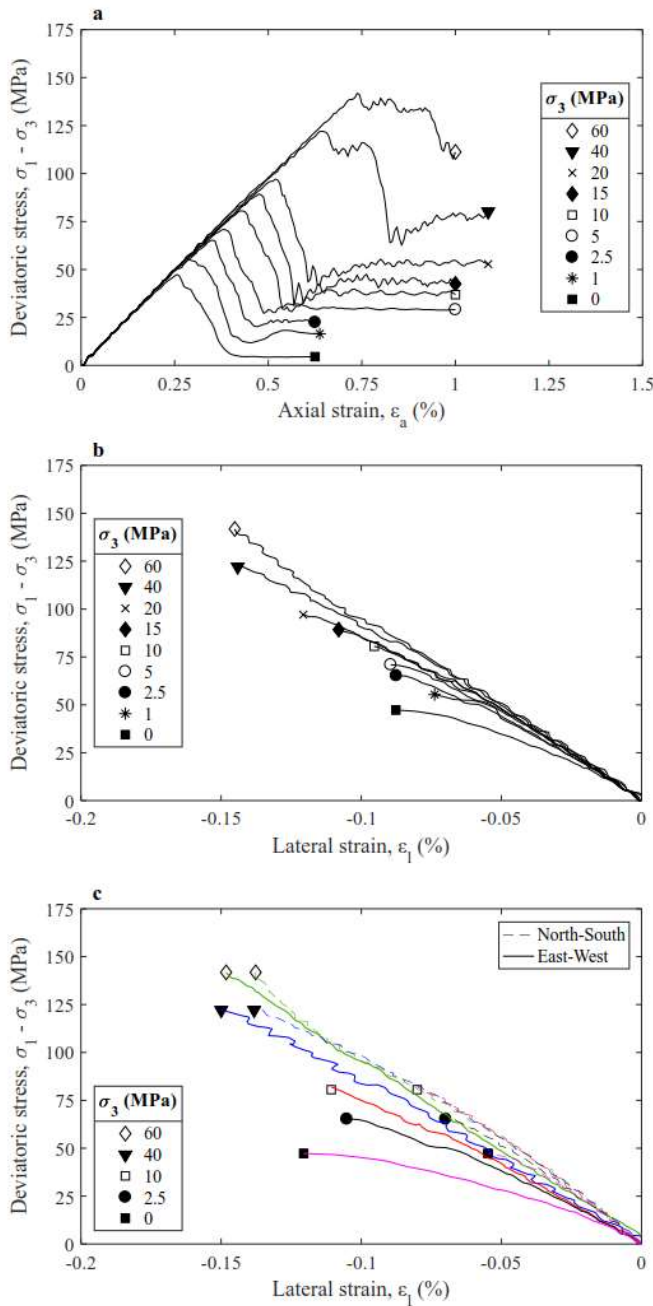


Figure 10. Results of simulated triaxial tests on veined rock mass. a) Deviatoric stress ($\sigma_1 - \sigma_3$) vs. axial strain (ϵ_a) under different confining stress (σ_3). b) Deviatoric stress ($\sigma_1 - \sigma_3$) vs. lateral strain (ϵ_l) under different confining stress (σ_3). c) Examples of lateral strain in two perpendicular directions.

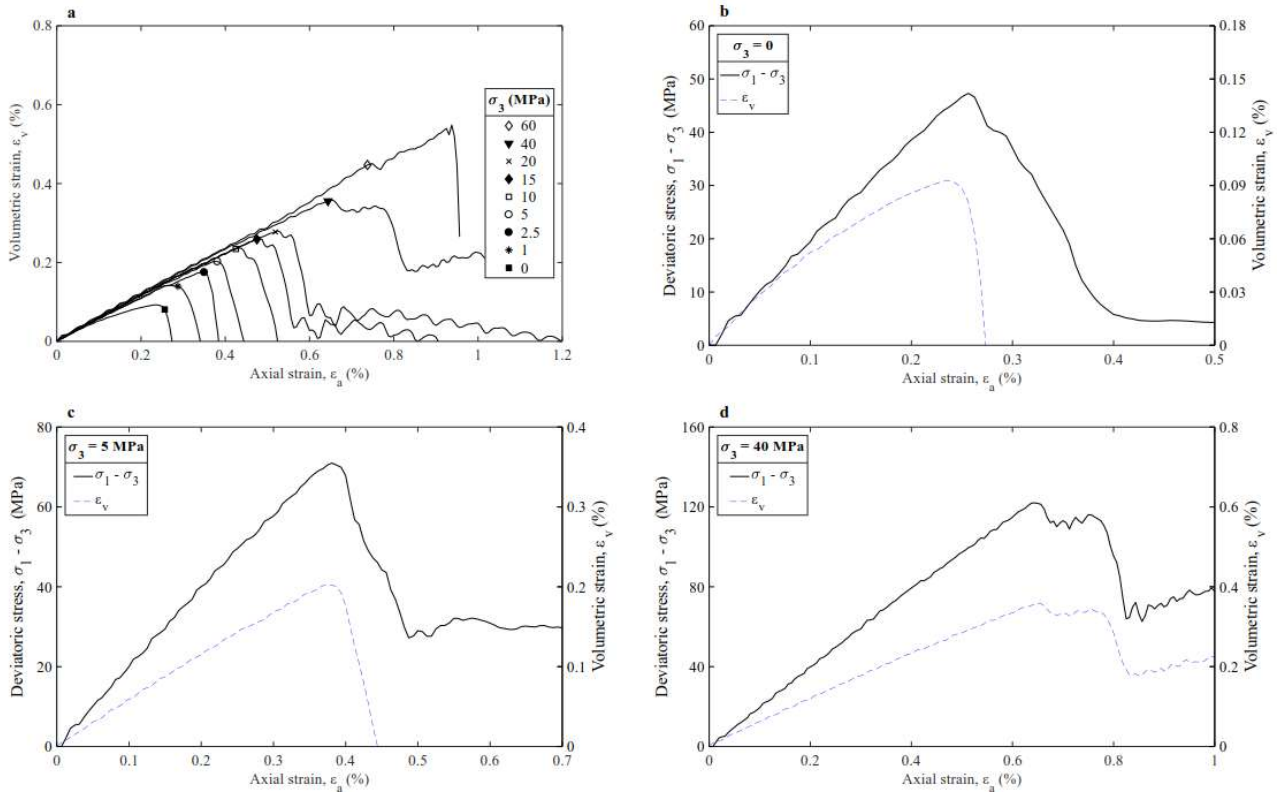


Figure 11. a) Volumetric strain (ϵ_v) vs. axial strain (ϵ_a). The symbol indicates the peak strength. b,c,d) Deviatoric stress ($\sigma_1 - \sigma_3$) and volumetric strain (ϵ_v) vs. lateral strain (ϵ_l) under different confining stresses (σ_3).

6.3.2 Influence of the intact rock strength on the rock mass strength

The strength of the intact rock is higher than that of the veins. At lower confinement, failure is expected to occur along the existing discontinuities rather than through the intact rock because of their lower strength. With increasing confining stress, failure through the intact rock can occur, because the failure becomes dominated by shear. To assess the influence of the adopted properties of the intact rock on the rock mass strength, test simulations considering the intact rock as linear elastic were also carried out.

The comparison of both simulations is shown in Fig. 12. It is observed that both models have the same uniaxial compressive strength. The difference in the strength increases with increasing confining stress, until both envelopes are almost parallel. This confirms the results on the modeled core-size specimens shown in Section 5, in which the failure in uniaxial compression occurs mainly through the veins.

The shape of the strength envelope is not affected by the strength of the intact rock. In both cases, with the rock as linear-elastic or as elasto-plastic material, the strength envelope is non-linear. The strength of the veins was modeled using the Mohr-Coulomb failure criterion, which has linear dependency on the confining stress. Their elastic behavior is linear. Consequently, the non-linearity of the strength envelope is produced by the interaction of the discontinuity network during failure.

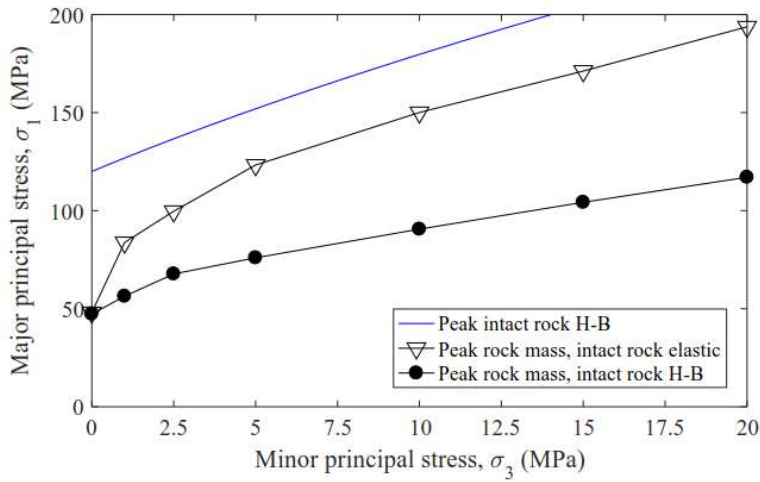


Figure 12. Peak strength of simulated triaxial tests on veined rock mass. Comparison between intact rock as linear elastic and as elasto-plastic material with Hoek-Brown failure criterion

6.3.3 Rock mass strength envelope

Because the primary copper ore at the El Teniente has almost no fractures, it is difficult to characterize it using, for example, the GSI or RMR to scale the laboratory results to the rock mass. The rock mass quality of the CMET lithology at El Teniente mine has been classified as good to very good, with a GSI in the range 70 to 90 and RMR between 66 and 74.²⁶ However, since the discontinuities correspond to filled and cemented veins, and not to fractures, the assumption of those values is not clear.

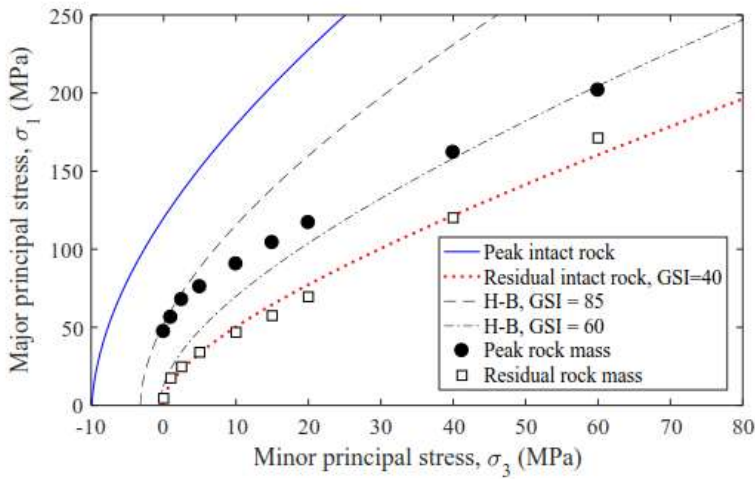


Figure 13. Results of simulated triaxial tests on veined rock mass and strength envelopes according to the Hoek-Brown failure criterion.

The failure envelope obtained from the simulations is shown in Fig.13. As mentioned above, the strength envelope is not linear and follows a curve that might be fit by the Hoek-Brown criterion. The envelopes of the Hoek-Brown criterion belonging to different GSI values are also shown in Fig.13. It is observed that the envelopes, calculated using Eq. 3, produce only an approximate fit to the data. A value of GSI=85 results in a good approximation to the rock mass strength at low confining stress. At high confining stress, the strength is better approximated assuming GSI=60. The best fit to the residual strength is achieved assuming GSI=40, which is also the residual GSI of the intact rock. This

seems to agree with similar observations for jointed rock masses, reported by Alejano et al.²³, Cai et al.²⁷ and Gao and Kang²⁸ who found that the residual strength of jointed rock masses may be similar to the residual strength of intact rock regardless of fracture intensity.

The uniaxial strength of the modeled rock mass block is significantly lower than that of the core-size specimens described in Section 5. This can be attributed to the vein density. In the rock mass the inclusion of more interconnected discontinuities, makes the veined rock mass weaker. Numerical simulations at considerable larger scale²⁰ and numerical back-analysis⁵ suggest lower rock mass strength at mine-scale. The simulations presented here consider a rock mass with veins but without open fractures. In a larger scale (tens of meters), the presence of fractures would additionally reduce the strength of the rock mass.

6.3.4 Elastic properties

The tangent deformation modulus and Poisson's ratio were calculated from the stress-strain curve at 50 % of the peak strength. The secant deformation modulus (E_s) was calculated between zero and the strain at peak strength. The obtained deformation modulus is about 20 GPa. The Poisson's ratio varies depending on the direction of measurement. In general, the east-west direction delivers greater Poisson's ratio. The results are summarized in Table 11.

The deformation modulus can be estimated indirectly from rock mass parameters and classification systems.²⁹⁻³³ Several equations found in the literature for the estimation of the rock mass modulus are presented in Table 12. Using these equations, the rock mass modulus was calculated assuming a GSI of 60 and 85. The RMR was estimated as $RMR = GSI + 5$.³⁴ An RMR of 65 and 90 is equivalent to a RMI of 5.7 and 147.²⁹

The best approximation to the rock mass modulus was obtained with the equation of Nicholson and Bieniawski³¹ and that of Palmstrom.² The other equations tend to overestimate the stiffness, even if those approaches do not consider cementation in the discontinuities, which should result in lower deformation modulus.

Table 11. Strength and elastic parameters of numerical model of veined rock mass

σ_3 (MPa)	$\sigma_1 - \sigma_3$ (MPa)	σ_1 (MPa)	σ_{1r} (MPa)	ν east-west	ν north-south	ν	E_t (GPa)	E_s (GPa)
0	47.3	47.3	4.6	0.44	0.19	0.32	19.2	18.5
1	55.4	56.4	17.5	0.3	0.17	0.24	19.5	19.3
2.5	65.3	67.8	24.8	0.27	0.2	0.24	19	18.6
5	71.0	76.0	34.1	0.27	0.16	0.22	19.8	18.6
10	80.6	90.6	46.9	0.24	0.18	0.21	19.2	19
15	89.3	104.3	57.5	0.24	0.21	0.23	20.4	18.8
20	97.1	117.1	69.5	0.24	0.22	0.23	20.7	18.7
40	122.0	162.0	120.2	0.23	0.21	0.22	20	19
60	141.8	201.8	171.2	0.2	0.25	0.22	19.9	19.2

Table 12. Estimates of the rock mass deformation modulus. The intact rock Young's modulus is $E_i = 55$ GPa and the damage factor is $D = 0$

Author	Equation	GSI/RMR/RMi	E_{rm} (GPa)
Bieniawski ³²	$E_{rm} = 2RMR - 100$	65 - 90	30 - 80
Nicholson and Bieniawski ³¹	$E_{rm} = \frac{E_i}{100} \left(0.0028RMR^2 + 0.9e^{\frac{RMR}{22.82}} \right)$	65 - 90	15 - 38
Serafim and Pereira ³⁰	$E_{rm} = 10^{\frac{RMR-10}{40}}$	65 - 90	24 - 100
Palmström ²	$E_{rm} = 5.6RMR^{0.375}$	5.7 - 147	11 - 36
Hoek and Diederichs ³³	$E_{rm} = E_i \left(0.02 + \frac{1 - D/2}{1 + e^{\frac{60+15D-GSI}{11}}} \right)$	60 - 85	29 - 51
Numerical model of veined rock mass	-	-	20

6.3.5 Failure mode

The failure mode was observed to depend on the applied confining stress. In uniaxial compression, the modeled block fails mainly by shearing and tension of two veins, forming a wedge as shown in Fig. 14a. Almost no failure takes place in the intact rock just before reaching the peak strength. In triaxial compression, under confining stress of 20 MPa, failure occurs along a different plane and through the intact rock (Fig. 14b).

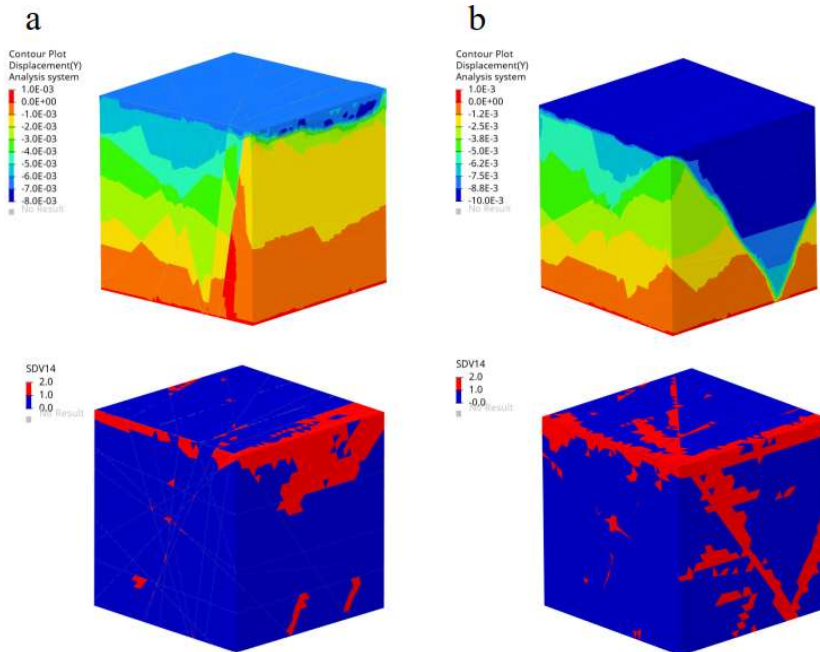


Figure 14. Results of simulated uniaxial (a) and triaxial test under $\sigma_3 = 20$ MPa (b). Vertical displacement and Elements in failure after reaching peak strength (red)

The difference in the failure mode can be observed in the cumulative number of elements reaching failure (tensile or shear). The results of the simulations under $\sigma_3 = 0, 5$ and 20 MPa are shown in Fig. 15. It is observed that the number of elements in failure after peak strength decreases with increasing confinement, indicating a more localized failure. This agrees with that normally expected

in a triaxial test, in which the failure changes with increasing confinement from spalling to shear along a plane.

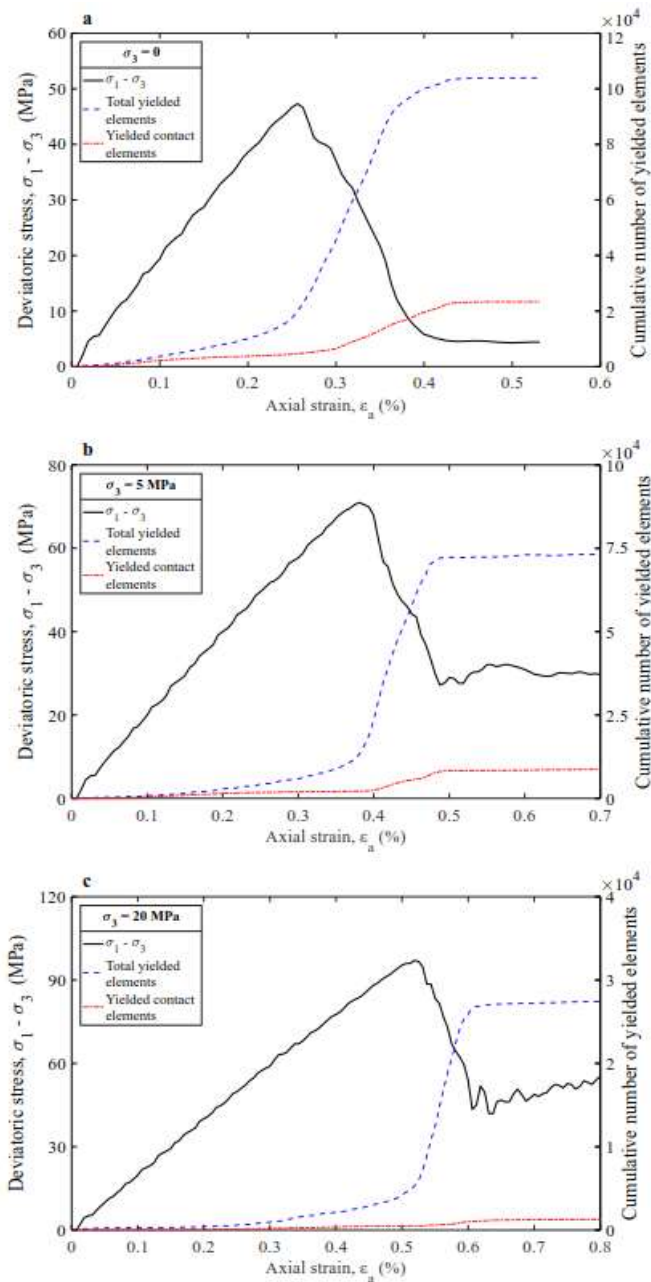


Figure 15. Deviatoric stress ($\sigma_1 - \sigma_3$) stress and cumulative number of elements in failure vs. axial strain (ϵ_a). a) Uniaxial compression. b) $\sigma_3 = 5$ MPa. c) $\sigma_3 = 20$ MPa

The fraction of contact elements in the total yielded elements decreases with increasing confining stress. The total number of vein elements in failure also decreases, indicating that failure of intact rock becomes more important and that less veins participate in the failure mechanism. This can be expected, since in uniaxial compression the failure initiates along the existing planes of weakness or veins (as shown in Section 5) and as the confining stress increases, failure includes the intact rock.

In the uniaxial test, the failure of the elements starts with loading, increasing at the peak and during softening. In triaxial compression, there is initially limited activity, with limited number of elements reaching failure. The failure increases significantly shortly before reaching the peak strength

(Fig. 15b and c). This is related to the onset of dilation, that, as mentioned before, occurs before reaching the peak strength in the uniaxial tests, whereas in triaxial compression occurs almost simultaneously when reaching the peak strength. It also indicates a more localized failure under confinement.

7 Conclusions

In the present work, the finite element method was used to study the behavior of veined rock mass. The veined rock was modeled as an assemblage of blocks representing the intact rock, divided and bonded by thin layers representing the veins. The applicability of the cohesive elements to simulate the rock veins was examined and validated. The advantage of the applied method is that the input parameters for the intact rock and discontinuities can be determined directly from laboratory tests without previous calibration.

Uniaxial tests on veined rock samples were simulated and the results compared to those of laboratory tests. The strength, elastic parameters and the failure mode obtained from the simulations showed good agreement with those obtained experimentally. The numerical model also reproduced the brittle failure observed in the tests. These results validate the capability of the model to reproduce the behavior of veined rock.

Using the same methodology, a veined rock mass volume of the CMET lithology from the El Teniente mine was modeled. Simulations of uniaxial and triaxial tests were performed on the veined rock mass block. In uniaxial compression, the behavior resulted less brittle than in the modeled core-size specimens, which can be attributed to the density of discontinuities interacting during failure. It was found that the strength of the intact rock only affected the global strength if a confining pressure was applied.

The obtained strength envelope of the veined rock mass is non-linear; however, the Hoek-Brown criterion could not be fit to the strength by varying the GSI. The strength envelope lies between the curves given by the Hoek-Brown criterion calculated with GSI values of 60 and 85. The curved shape of the strength envelope showed to be independent of the intact rock strength, which can be attributed to the interaction of the discontinuities during failure.

The residual strength of the veined rock coincides with the residual strength of the intact rock, which agrees with previously reported observations in jointed rock.

The Young's and deformation moduli of the modeled veined rock mass was about 20 GPa. With the range of GSI obtained for the strength envelope, the rock deformation modulus was calculated using different equations available in the literature. Some of these equations deliver values that are comparable to the obtained stiffness. Nevertheless, these estimations are obtained assuming classification and rating values that cannot be directly applied in the veined rock mass.

The failure mode was observed to change with the confining stress. The results suggest more localization of the failure within the rock mass with increasing confinement

Further investigation is required to better understand some of the findings of this study, such as the influence of strength and stiffness of the intact rock and veins in the behavior of the veined rock mass.

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References

- [1] Bieniawski ZT. Engineering classification of rock masses. *Trans S African Inst Civ Engrs.* 1973;15(12):335–44
- [2] Palmström A. Characterizing rock masses by the RMi for use in practical rock engineering: Part 1: The development of the Rock Mass index (RMi). *Tunnelling and underground space technology.* 1996;11(2):175-188
- [3] Hoek E, Brown ET. Practical estimates of rock mass strength. *Int. J. Rock Mech. Min. Sci. and Geomech. Abstr.* 1997;34 (8):1165-1186
- [4] Barton N, Grimstad E. The Q-system following twenty years of application in NMT support selection. In: *Proceedings of 43rd Geomechanic Colloquium, Salzburg.* 1994. p. 428.
- [5] Pardo C, Villaescusa E, Beck D, Brzovic A. Back Analysis of intensive rock mass damage at the El Teniente Mine. In: *Proceedings of the CRC-Mining Conference, Brisbane.* 2012
- [6] Rojas E, Cavieres P, Dunlop R, Gaete S. Control of induced seismicity at the El Teniente mine, Codelco Chile. In: *Proceedings of the Massmin 2000. Brisbane.* 29 October-2 November 2000. p. 777-781
- [7] Brzovic A, Herrera S. Assessing geological vein size and intensity using Discrete Fracture Network modelling at the El Teniente Mine, Chile. In: *Proceedings of the 45th US Rock Mechanics/Geomechanics Symposium.* San Francisco. 26–29, June 2011.
- [8] Brzovic A, Villaescusa E. Rock mass characterization and assessment of block-forming geological discontinuities during caving of primary copper ore at the El Teniente mine, Chile. *Int. J. Rock Mech. Min. Sci. and Geomech. Abstr.* 2007;44:565-583
- [9] Lee HP, Olson JE, Holder J, Gale JF, Myers RD. The interaction of propagating opening mode fractures with preexisting discontinuities in shale. *Journal of Geophysical Research: Solid Earth.* 2015;120(1):169-181.
- [10] Turichshev A, Hadjigeorgiou J. Experimental and numerical investigations into the strength of intact veined rock. *Rock Mechanics and Rock Engineering.* 2015;48(5):1897-1912
- [11] Turichshev A, Hadjigeorgiou J. Triaxial compression experiments on intact veined andesite. *International Journal of Rock Mechanics and Mining Sciences.* 2016;86:179-193
- [12] Vallejos J, Suzuki K, Brzovic A, Mas Ivars D. Application of the Synthetic Rock Mass modeling to veined core-size samples. *Int. J. Rock Mech. Min. Sci.* 2016;81:47-61
- [13] Itasca Consulting Group. PFC3D – Particle Flow Code in 3 Dimensions. Minneapolis, USA
- [14] Shang J. rupture of veined granite in polyaxial compression: Insights From Three-Dimensional Discrete Element Method Modeling. *Journal of Geophysical Research: Solid Earth.* 2020;125(2):1-25
- [15] Bai Q, Young RP. Numerical investigation of the mechanical and damage behaviors of veined gneiss during true-triaxial stress path loading by simulation of in situ conditions. *Rock Mechanics and Rock Engineering.* 2020;53:133-151.
- [16] Virgo S, Abe S, Urai JL. Extension fracture propagation in rocks with veins: Insight into the crack-seal process using Discrete Element Method modeling. *Journal of Geophysical Research: Solid Earth.* 2013;118(10):5236-5251.
- [17] Suzuki K. Application of Synthetic Rock Mass in laboratory scale (in Spanish). MSc thesis. Universidad de Chile, Santiago, Chile; 2013.

- [18] Dassault Systemes. ABAQUS-Finite element analysis. Version 6.18. Vélizy-Villacoublay, France.
- [19] Jacobsson L, Flansbjer M, Brzovic A, de los Santos C. Direct shear and tensile test on cemented healed joints from El Teniente Mine, Chile. In: Proceedings of the 12th ISRM Congress. Beijing. 2011.
- [20] Vallejos J, Suzuki K, Brzovic A and Mas Ivars D. Characterization and synthetic simulations to determine rock mass behaviour at the El Teniente mine, Chile. Part II. In: Proceedings of the 3rd International Symposium on Block and Sublevel Caving. Santiago; 5-6 June 2014.
- [21] MATLAB and Statistics Toolbox Release 2014, The MathWorks, Inc., Natick, Massachusetts, United States
- [22] Eberhardt E. The Hoek-Brown failure criterion. Rock mechanics and rock engineering. 2012;45(6):981-988.
- [23] Alejano LR, Arzúa J, Bozorgzadeh N, Harrison JP. Triaxial strength and deformability of intact and increasingly jointed granite samples. International Journal of Rock Mechanics and Mining Sciences. 2017;95:87-103.
- [24] Kulatilake PHSW, Ucpirti H, Wang S, Radberg G, Stephansson O. Use of the distinct element method to perform stress analysis in rock with non-persistent joints and to study the effect of joint geometry parameters on the strength and deformability of rock masses. Rock Mech Rock Eng. 1992;25(4):253-274
- [25] Golder. FracMan Technology Group. Available: www.fracman.golder.com
- [26] Brzovic A. Fundamentos geológicos para un sistema de clasificación geotécnico del macizo rocoso primario, mina El Teniente. Internal report SGL-87/2001 of the Superintendence Geology, CODELCO-Chile Division El Teniente. 2001.
- [27] Cai M, Kaiser PK, Tasaka Y, Minamic M. Determination of residual strength parameters of jointed rock masses using the GSI system. Int J Rock Mech Min Sci. 2007;44:247-265.
- [28] Gao FQ, Kang HP. Effects of pre-existing discontinuities on the residual strength of rock mass – insight from a discrete element method simulation. J Struct Geol. 2016;85:40-50.
- [29] Palmström A. Combining the RMR, Q, and RMi classification systems. Tunnelling and Underground Space Technology. 2009;24(4):491-492
- [30] Serafim JL, Pereira JP. Consideration of the geomechanical classification of Bieniawski. In: Proceedings of the International Symposium on Engineering Geology and Underground Construction. Lisbon. 1983. p. 33-44
- [31] Nicholson GA, Bieniawski ZT. A nonlinear deformation modulus based on rock mass classification. International journal of Mining and geological engineering. 1990;8:181-202
- [32] Bieniawski ZT. Determining rock mass deformability-experience from case histories. Int. J. Rock Mech. Min. Sci. and Geomech. Abstr. 1978;15(5):237-247
- [33] Hoek E, Diederichs MS. Empirical estimation of rock mass modulus. Int. J. Rock Mech. Min. Sci. and Geomech. Abstr. 2006;43(2):203-215
- [34] Ceballos F, Olalla C, Jiménez R. Relationship between RMRb and GSI based on in situ data. In: Proceedings of ISRM Regional Symposium-EUROCK 2014. Vigo; 27-29 May 2014.